In [ ]:
```
EXPERIMENT - 4
DATA MANIPULATION WITH PANDAS
PIN Number: 241UDS0103
Aim: To perform data manipulation operations using Pandas on a new eco-
retail dataset.
```

In [ ]:
```
### Step 1 – Install and Import Pandas Library
```

In [1]:
```
import pandas as pd
```

In [ ]:
```
### Step 2 – Importing the Dataset
```

In [2]:
```
df_241UDS0103 = pd.read_csv("Experiment_4_241UDS0103_Eco_Retail_Data.csv")
df_241UDS0103.head(5)
```

Out[2]: Columns 1–8 of 11

| | Item_ID | Item_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent |
|---|---|---|---|---|---|---|---|
| **0** | E201 | Solar Lantern | Renewable Energy | Vijayawada | 38 | 850 | 5.0 |
| **1** | E202 | Smart Thermostat | Smart Home | Chennai | 27 | 2400 | 8.0 |
| **2** | E203 | Water Purifier | Home Appliance | Visakhapatnam | 44 | 3200 | 10.0 |
| **3** | E204 | Portable Charger | Electronics | Hyderabad | 63 | 1250 | 7.0 |
| **4** | E205 | Air Quality Sensor | Smart Home | Pune | 21 | 1850 | 6.0 |

Columns 9–11 of 11 · first column repeated

| | Customer_Rating | Stock_Available | Payment_Method |
|---|---|---|---|
| **0** | 4.4 | 24 | UPI |
| **1** | 4.6 | 15 | Card |
| **2** | 4.2 | 19 | Cash |
| **3** | 4.5 | 37 | UPI |
| **4** | 4.7 | 12 | Card |

In [ ]:
```
### Step 3 – Set Index for the DataFrame
```

In [3]:
```
df_241UDS0103 = df_241UDS0103.set_index("Item_ID")
df_241UDS0103.head()
```

Out[3]: Columns 1–8 of 10

| Item_ID | Item_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent | Customer_Rating |
|---|---|---|---|---|---|---|---|
| **E201** | Solar Lantern | Renewable Energy | Vijayawada | 38 | 850 | 5.0 | 4.4 |
| **E202** | Smart Thermostat | Smart Home | Chennai | 27 | 2400 | 8.0 | 4.6 |
| **E203** | Water Purifier | Home Appliance | Visakhapatnam | 44 | 3200 | 10.0 | 4.2 |
| **E204** | Portable Charger | Electronics | Hyderabad | 63 | 1250 | 7.0 | 4.5 |
| **E205** | Air Quality Sensor | Smart Home | Pune | 21 | 1850 | 6.0 | 4.7 |

Columns 9–10 of 10 · first column repeated

| Item_ID | Stock_Available | Payment_Method |
|---|---|---|
| **E201** | 24 | UPI |
| **E202** | 15 | Card |
| **E203** | 19 | Cash |
| **E204** | 37 | UPI |
| **E205** | 12 | Card |

In [ ]:
```
### Step 4 – Info of the DataFrame
```

In [4]:
```
df_241UDS0103.info()
```

```
<class 'pandas.DataFrame'>
Index: 30 entries, E201 to E230
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   Item_Name         30 non-null     str    
 1   Category          30 non-null     str    
 2   City              30 non-null     str    
 3   Units_Sold        30 non-null     int64  
 4   Unit_Price        30 non-null     int64  
 5   Discount_Percent  30 non-null     float64
 6   Customer_Rating   30 non-null     float64
 7   Stock_Available   30 non-null     int64  
 8   Payment_Method    30 non-null     str    
dtypes: float64(2), int64(3), str(4)
memory usage: 2.3+ KB
```

In [ ]:
```
### Step 5 — Columns & Rows Selection
```

In [5]:
```
df_241UDS0103["Item_Name"]
```

Out[5]:
```
Item_ID
E201          Solar Lantern
E202       Smart Thermostat
E203         Water Purifier
E204       Portable Charger
E205     Air Quality Sensor
E206         LED Desk Light
E207            Compost Bin
E208           Bamboo Shelf
E209        Reusable Bottle
E210      Induction Cooktop
E211      Wireless Doorbell
E212         Garden Sprayer
E213           Laptop Stand
E214        Ceramic Planter
E215         Power Inverter
E216    Insulated Lunch Jar
E217              Air Fryer
E218       Solar Wall Light
E219         Foldable Stool
E220            Noise Meter
E221        Electric Kettle
E222      Recycled Notebook
E223          Motion Sensor
E224       Travel Organizer
E225         Mini Projector
E226                Desk Mat
E227            Rain Barrel
E228       Solar Power Bank
E229         Storage Basket
E230                USB Hub
Name: Item_Name, dtype: str
```

In [ ]:
```
### Step 6 — Selecting Multiple Columns
```

In [6]:
```
df_241UDS0103[["Item_Name", "Category", "City"]].head(3)
```

Out[6]:

| Item_ID | Item_Name | Category | City |
|---|---|---|---|
| **E201** | Solar Lantern | Renewable Energy | Vijayawada |
| **E202** | Smart Thermostat | Smart Home | Chennai |
| **E203** | Water Purifier | Home Appliance | Visakhapatnam |

In [ ]:
```
### Step 7 — Select Row Using iloc
```

In [7]:
```
df_241UDS0103.iloc[1]
```

Out[7]:
```
Item_Name           Smart Thermostat
Category                  Smart Home
City                         Chennai
Units_Sold                        27
Unit_Price                      2400
Discount_Percent                 8.0
Customer_Rating                  4.6
Stock_Available                   15
Payment_Method                  Card
Name: E202, dtype: object
```

In [ ]:
```
### Step 8 — Select Row Using loc
```

In [8]:
```
df_241UDS0103.loc["E230"]
```

Out[8]:
```
Item_Name              USB Hub
Category           Electronics
City                 Bengaluru
Units_Sold                  66
Unit_Price                 570
Discount_Percent          10.0
Customer_Rating            4.5
Stock_Available             40
Payment_Method            Card
Name: E230, dtype: object
```

In [ ]:
```
### Step 9 — Manipulating Data Based on Columns
```

In [9]:
```
df_241UDS0103.head(2)
```

Out[9]:

Columns 1–8 of 10

| | Item_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent | Customer_Rating |
|---|---|---|---|---|---|---|---|
| **Item_ID** | | | | | | | |
| **E201** | Solar Lantern | Renewable Energy | Vijayawada | 38 | 850 | 5.0 | 4.4 |
| **E202** | Smart Thermostat | Smart Home | Chennai | 27 | 2400 | 8.0 | 4.6 |

Columns 9–10 of 10 · first column repeated

| | Stock_Available | Payment_Method |
|---|---|---|
| **Item_ID** | | |
| **E201** | 24 | UPI |
| **E202** | 15 | Card |

In [ ]:
```
### Step 10 — Creating Total Sales Column
```

In [10]:
```
df_241UDS0103["Total_Sales_241UDS0103"] = df_241UDS0103["Units_Sold"] *
df_241UDS0103["Unit_Price"]
df_241UDS0103.head(2)
```

Out[10]: Columns 1–8 of 11

| | Item_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent | Customer_Rating |
|---|---|---|---|---|---|---|---|
| Item_ID | | | | | | | |
| E201 | Solar Lantern | Renewable Energy | Vijayawada | 38 | 850 | 5.0 | 4.4 |
| E202 | Smart Thermostat | Smart Home | Chennai | 27 | 2400 | 8.0 | 4.6 |

Columns 9–11 of 11 · first column repeated

| | Stock_Available | Payment_Method | Total_Sales_241UDS0103 |
|---|---|---|---|
| Item_ID | | | |
| E201 | 24 | UPI | 32300 |
| E202 | 15 | Card | 64800 |

In [ ]:
```
### Step 11 — Creating Discount Amount Column
```

In [11]:
```
df_241UDS0103["Discount_Amount_241UDS0103"] =
df_241UDS0103["Total_Sales_241UDS0103"] * df_241UDS0103["Discount_Percent"]
/ 100
df_241UDS0103.head(2)
```

Out[11]: Columns 1–8 of 12

| | Item_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent | Customer_Rating |
|---|---|---|---|---|---|---|---|
| Item_ID | | | | | | | |
| E201 | Solar Lantern | Renewable Energy | Vijayawada | 38 | 850 | 5.0 | 4.4 |
| E202 | Smart Thermostat | Smart Home | Chennai | 27 | 2400 | 8.0 | 4.6 |

Columns 9–12 of 12 · first column repeated

| | Stock_Available | Payment_Method | Total_Sales_241UDS0103 | Discount_Amount_241UDS0103 |
|---|---|---|---|---|
| Item_ID | | | | |
| E201 | 24 | UPI | 32300 | 1615.0 |
| E202 | 15 | Card | 64800 | 5184.0 |

In [ ]:
```
### Step 12 — Creating Net Sales Column
```

In [12]:
```
df_241UDS0103["Net_Sales_241UDS0103"] =
df_241UDS0103["Total_Sales_241UDS0103"] -
df_241UDS0103["Discount_Amount_241UDS0103"]
df_241UDS0103.head(2)
```

Out[12]: Columns 1–8 of 13

| Item_ID | Item_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent | Customer_Rating |
|---|---|---|---|---|---|---|---|
| **E201** | Solar Lantern | Renewable Energy | Vijayawada | 38 | 850 | 5.0 | 4.4 |
| **E202** | Smart Thermostat | Smart Home | Chennai | 27 | 2400 | 8.0 | 4.6 |

Columns 9–13 of 13 · first column repeated

| Item_ID | Stock_Available | Payment_Method | Total_Sales_241UDS0103 | Discount_Amount_241UDS0103 | Net_Sales_241UDS0103 |
|---|---|---|---|---|---|
| **E201** | 24 | UPI | 32300 | 1615.0 | 30685.0 |
| **E202** | 15 | Card | 64800 | 5184.0 | 59616.0 |

In [ ]:
```
### Step 13 — Round Calculated Values
```

In [13]:
```
df_241UDS0103["Discount_Amount_241UDS0103"] =
df_241UDS0103["Discount_Amount_241UDS0103"].round(2)
df_241UDS0103["Net_Sales_241UDS0103"] =
df_241UDS0103["Net_Sales_241UDS0103"].round(2)
df_241UDS0103.head(2)
```

Out[13]: Columns 1–8 of 13

| Item_ID | Item_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent | Customer_Rating |
|---|---|---|---|---|---|---|---|
| **E201** | Solar Lantern | Renewable Energy | Vijayawada | 38 | 850 | 5.0 | 4.4 |
| **E202** | Smart Thermostat | Smart Home | Chennai | 27 | 2400 | 8.0 | 4.6 |

Columns 9–13 of 13 · first column repeated

| Item_ID | Stock_Available | Payment_Method | Total_Sales_241UDS0103 | Discount_Amount_241UDS0103 | Net_Sales_241UDS0103 |
|---|---|---|---|---|---|
| **E201** | 24 | UPI | 32300 | 1615.0 | 30685.0 |
| **E202** | 15 | Card | 64800 | 5184.0 | 59616.0 |

In [ ]:
```
### Step 14 – Conditional Filtering
```

In [14]:
```
df_241UDS0103[df_241UDS0103["Units_Sold"] > 50].head(2)
```

Out[14]: Columns 1–8 of 13

| Item_ID | Item_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent | Customer_Rating |
|---|---|---|---|---|---|---|---|
| **E204** | Portable Charger | Electronics | Hyderabad | 63 | 1250 | 7.0 | 4.5 |
| **E206** | LED Desk Light | Electronics | Mumbai | 72 | 680 | 12.0 | 4.1 |

Columns 9–13 of 13 · first column repeated

| Item_ID | Stock_Available | Payment_Method | Total_Sales_241UDS0103 | Discount_Amount_241UDS0103 | Net_Sales_241UDS0103 |
|---|---|---|---|---|---|
| **E204** | 37 | UPI | 78750 | 5512.5 | 73237.5 |
| **E206** | 43 | Cash | 48960 | 5875.2 | 43084.8 |

In [ ]:
```
### Step 15 – Products with High Sales AND Low Stock
```

In [15]:
```
df_241UDS0103[(df_241UDS0103["Total_Sales_241UDS0103"] > 30000) &
(df_241UDS0103["Stock_Available"] < 25)].head(3)
```

Out[15]: Columns 1–8 of 13

| | Item_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent | Customer_Rating |
|---|---|---|---|---|---|---|---|
| Item_ID | | | | | | | |
| E201 | Solar Lantern | Renewable Energy | Vijayawada | 38 | 850 | 5.0 | 4.4 |
| E202 | Smart Thermostat | Smart Home | Chennai | 27 | 2400 | 8.0 | 4.6 |
| E203 | Water Purifier | Home Appliance | Visakhapatnam | 44 | 3200 | 10.0 | 4.2 |

Columns 9–13 of 13 · first column repeated

| | Stock_Available | Payment_Method | Total_Sales_241UDS0103 | Discount_Amount_241UDS0103 | Net_Sales_241UDS0103 |
|---|---|---|---|---|---|
| Item_ID | | | | | |
| E201 | 24 | UPI | 32300 | 1615.0 | 30685.0 |
| E202 | 15 | Card | 64800 | 5184.0 | 59616.0 |
| E203 | 19 | Cash | 140800 | 14080.0 | 126720.0 |

In [ ]:
```
### Step 16 – Products from Hyderabad OR Mumbai
```

In [16]:
```
df_241UDS0103[(df_241UDS0103["City"] == "Hyderabad") |
(df_241UDS0103["City"] == "Mumbai")].head()
```

Out[16]: Columns 1–8 of 13

| Item_ID | Item_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent | Customer_Rating |
|---|---|---|---|---|---|---|---|
| **E204** | Portable Charger | Electronics | Hyderabad | 63 | 1250 | 7.0 | 4.5 |
| **E206** | LED Desk Light | Electronics | Mumbai | 72 | 680 | 12.0 | 4.1 |
| **E209** | Reusable Bottle | Lifestyle | Hyderabad | 56 | 420 | 10.0 | 4.8 |
| **E210** | Induction Cooktop | Home Appliance | Mumbai | 25 | 3100 | 11.0 | 4.4 |
| **E217** | Air Fryer | Home Appliance | Hyderabad | 29 | 3600 | 12.0 | 4.6 |

Columns 9–13 of 13 · first column repeated

| Item_ID | Stock_Available | Payment_Method | Total_Sales_241UDS0103 | Discount_Amount_241UDS0103 | Net_Sales_241UDS0103 |
|---|---|---|---|---|---|
| **E204** | 37 | UPI | 78750 | 5512.5 | 73237.5 |
| **E206** | 43 | Cash | 48960 | 5875.2 | 43084.8 |
| **E209** | 29 | UPI | 23520 | 2352.0 | 21168.0 |
| **E210** | 16 | Card | 77500 | 8525.0 | 68975.0 |
| **E217** | 17 | Card | 104400 | 12528.0 | 91872.0 |

In [ ]:
```
### Step 17 — Products with Rating ≥ 4.5 AND Stock < 30
```

In [17]:

```
df_241UDS0103[(df_241UDS0103["Customer_Rating"] ⩾ 4.5) &
(df_241UDS0103["Stock_Available"] < 30)]
```

Out[17]: Columns 1–8 of 13

| Item_ID | Item_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent | Customer_Rating |
|---|---|---|---|---|---|---|---|
| **E202** | Smart Thermostat | Smart Home | Chennai | 27 | 2400 | 8.0 | 4.6 |
| **E205** | Air Quality Sensor | Smart Home | Pune | 21 | 1850 | 6.0 | 4.7 |
| **E209** | Reusable Bottle | Lifestyle | Hyderabad | 56 | 420 | 10.0 | 4.8 |
| **E212** | Garden Sprayer | Garden | Pune | 39 | 760 | 8.0 | 4.5 |
| **E213** | Laptop Stand | Accessories | Chennai | 31 | 1150 | 6.0 | 4.6 |
| **E215** | Power Inverter | Electronics | Bengaluru | 22 | 4200 | 9.0 | 4.7 |
| **E217** | Air Fryer | Home Appliance | Hyderabad | 29 | 3600 | 12.0 | 4.6 |
| **E218** | Solar Wall Light | Renewable Energy | Delhi | 53 | 1150 | 10.0 | 4.5 |
| **E220** | Noise Meter | Smart Home | Chennai | 16 | 2250 | 8.0 | 4.8 |
| **E223** | Motion Sensor | Smart Home | Visakhapatnam | 32 | 1350 | 11.0 | 4.7 |
| **E224** | Travel Organizer | Accessories | Delhi | 45 | 690 | 9.0 | 4.5 |
| **E225** | Mini Projector | Electronics | Mumbai | 19 | 5400 | 14.0 | 4.6 |
| **E228** | Solar Power Bank | Renewable Energy | Chennai | 37 | 1750 | 12.0 | 4.8 |

Columns 9–13 of 13 · first column repeated

| Item_ID | Stock_Available | Payment_Method | Total_Sales_241UDS0103 | Discount_Amount_241UDS0103 | Net_Sales_241UDS0103 |
|---|---|---|---|---|---|
| **E202** | 15 | Card | 64800 | 5184.0 | 59616.0 |
| **E205** | 12 | Card | 38850 | 2331.0 | 36519.0 |
| **E209** | 29 | UPI | 23520 | 2352.0 | 21168.0 |
| **E212** | 20 | UPI | 29640 | 2371.2 | 27268.8 |
| **E213** | 18 | Card | 35650 | 2139.0 | 33511.0 |
| **E215** | 14 | Card | 92400 | 8316.0 | 84084.0 |
| **E217** | 17 | Card | 104400 | 12528.0 | 91872.0 |
| **E218** | 25 | UPI | 60950 | 6095.0 | 54855.0 |
| **E220** | 9 | Card | 36000 | 2880.0 | 33120.0 |
| **E223** | 13 | Card | 43200 | 4752.0 | 38448.0 |

| Item_ID | Stock_Available | Payment_Method | Total_Sales_241UDS0103 | Discount_Amount_241UDS0103 | Net_Sales_241UDS0103 |
|---|---|---|---|---|---|
| E224 | 23 | Cash | 31050 | 2794.5 | 28255.5 |
| E225 | 10 | UPI | 102600 | 14364.0 | 88236.0 |
| E228 | 19 | UPI | 64750 | 7770.0 | 56980.0 |

In [ ]:
```
### Step 18 – Filtering Multiple Cities Using .isin()
```

In [18]:
```
df_241UDS0103[df_241UDS0103["City"].isin(["Hyderabad", "Bengaluru",
"Pune"])].head()
```

Out[18]: Columns 1–8 of 13

| Item_ID | Item_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent | Customer_Rating |
|---|---|---|---|---|---|---|---|
| E204 | Portable Charger | Electronics | Hyderabad | 63 | 1250 | 7.0 | 4.5 |
| E205 | Air Quality Sensor | Smart Home | Pune | 21 | 1850 | 6.0 | 4.7 |
| E207 | Compost Bin | Garden | Bengaluru | 34 | 1450 | 9.0 | 4.3 |
| E209 | Reusable Bottle | Lifestyle | Hyderabad | 56 | 420 | 10.0 | 4.8 |
| E212 | Garden Sprayer | Garden | Pune | 39 | 760 | 8.0 | 4.5 |

Columns 9–13 of 13 · first column repeated

| Item_ID | Stock_Available | Payment_Method | Total_Sales_241UDS0103 | Discount_Amount_241UDS0103 | Net_Sales_241UDS0103 |
|---|---|---|---|---|---|
| E204 | 37 | UPI | 78750 | 5512.5 | 73237.5 |
| E205 | 12 | Card | 38850 | 2331.0 | 36519.0 |
| E207 | 22 | Card | 49300 | 4437.0 | 44863.0 |
| E209 | 29 | UPI | 23520 | 2352.0 | 21168.0 |
| E212 | 20 | UPI | 29640 | 2371.2 | 27268.8 |

In [ ]:
```
### Step 19 – Filtering Multiple Categories Using .isin()
```

In [19]:
```
df_241UDS0103[df_241UDS0103["Category"].isin(["Electronics", "Home Appliance"])].head()
```

Out[19]: Columns 1–8 of 13

| Item_ID | Item_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent | Customer_Rating |
|---|---|---|---|---|---|---|---|
| **E203** | Water Purifier | Home Appliance | Visakhapatnam | 44 | 3200 | 10.0 | 4.2 |
| **E204** | Portable Charger | Electronics | Hyderabad | 63 | 1250 | 7.0 | 4.5 |
| **E206** | LED Desk Light | Electronics | Mumbai | 72 | 680 | 12.0 | 4.1 |
| **E210** | Induction Cooktop | Home Appliance | Mumbai | 25 | 3100 | 11.0 | 4.4 |
| **E215** | Power Inverter | Electronics | Bengaluru | 22 | 4200 | 9.0 | 4.7 |

Columns 9–13 of 13 · first column repeated

| Item_ID | Stock_Available | Payment_Method | Total_Sales_241UDS0103 | Discount_Amount_241UDS0103 | Net_Sales_241UDS0103 |
|---|---|---|---|---|---|
| **E203** | 19 | Cash | 140800 | 14080.0 | 126720.0 |
| **E204** | 37 | UPI | 78750 | 5512.5 | 73237.5 |
| **E206** | 43 | Cash | 48960 | 5875.2 | 43084.8 |
| **E210** | 16 | Card | 77500 | 8525.0 | 68975.0 |
| **E215** | 14 | Card | 92400 | 8316.0 | 84084.0 |

In [ ]:
```
### Step 20 – Filtering Multiple Payment Methods Using .isin()
```

In [20]:
```
df_241UDS0103[df_241UDS0103["Payment_Method"].isin(["UPI", "Card"])].head()
```

Out[20]:

Columns 1–8 of 13

| Item_ID | Item_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent | Customer_Rating |
|---|---|---|---|---|---|---|---|
| **E201** | Solar Lantern | Renewable Energy | Vijayawada | 38 | 850 | 5.0 | 4.4 |
| **E202** | Smart Thermostat | Smart Home | Chennai | 27 | 2400 | 8.0 | 4.6 |
| **E204** | Portable Charger | Electronics | Hyderabad | 63 | 1250 | 7.0 | 4.5 |
| **E205** | Air Quality Sensor | Smart Home | Pune | 21 | 1850 | 6.0 | 4.7 |
| **E207** | Compost Bin | Garden | Bengaluru | 34 | 1450 | 9.0 | 4.3 |

Columns 9–13 of 13 · first column repeated

| Item_ID | Stock_Available | Payment_Method | Total_Sales_241UDS0103 | Discount_Amount_241UDS0103 | Net_Sales_241UDS0103 |
|---|---|---|---|---|---|
| **E201** | 24 | UPI | 32300 | 1615.0 | 30685.0 |
| **E202** | 15 | Card | 64800 | 5184.0 | 59616.0 |
| **E204** | 37 | UPI | 78750 | 5512.5 | 73237.5 |
| **E205** | 12 | Card | 38850 | 2331.0 | 36519.0 |
| **E207** | 22 | Card | 49300 | 4437.0 | 44863.0 |

In [ ]:
```
### Step 21 — Sorting Table
```

In [21]:
```
df_241UDS0103.sort_values("Units_Sold", ascending=False).head()
```

Out[21]:

Columns 1–8 of 13

| | Item_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent | Customer_Rating |
|---|---|---|---|---|---|---|---|
| **Item_ID** | | | | | | | |
| **E222** | Recycled Notebook | Stationery | Bengaluru | 84 | 160 | 5.0 | 4.4 |
| **E206** | LED Desk Light | Electronics | Mumbai | 72 | 680 | 12.0 | 4.1 |
| **E230** | USB Hub | Electronics | Bengaluru | 66 | 570 | 10.0 | 4.5 |
| **E204** | Portable Charger | Electronics | Hyderabad | 63 | 1250 | 7.0 | 4.5 |
| **E221** | Electric Kettle | Home Appliance | Pune | 61 | 920 | 6.0 | 4.2 |

Columns 9–13 of 13 · first column repeated

| | Stock_Available | Payment_Method | Total_Sales_241UDS0103 | Discount_Amount_241UDS0103 | Net_Sales_241UDS0103 |
|---|---|---|---|---|---|
| **Item_ID** | | | | | |
| **E222** | 52 | Cash | 13440 | 672.0 | 12768.0 |
| **E206** | 43 | Cash | 48960 | 5875.2 | 43084.8 |
| **E230** | 40 | Card | 37620 | 3762.0 | 33858.0 |
| **E204** | 37 | UPI | 78750 | 5512.5 | 73237.5 |
| **E221** | 38 | UPI | 56120 | 3367.2 | 52752.8 |

In [ ]:
```
### Step 22 — Sort by Unit Price — Ascending
```

In [22]:
```
df_241UDS0103.sort_values("Unit_Price", ascending=True).head()
```

Out[22]:

Columns 1–8 of 13

| | Item_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent | Customer_Rating |
|---|---|---|---|---|---|---|---|
| **Item_ID** | | | | | | | |
| **E222** | Recycled Notebook | Stationery | Bengaluru | 84 | 160 | 5.0 | 4.4 |
| **E226** | Desk Mat | Accessories | Vijayawada | 57 | 390 | 8.0 | 4.1 |
| **E209** | Reusable Bottle | Lifestyle | Hyderabad | 56 | 420 | 10.0 | 4.8 |
| **E214** | Ceramic Planter | Garden | Visakhapatnam | 47 | 540 | 13.0 | 4.1 |
| **E230** | USB Hub | Electronics | Bengaluru | 66 | 570 | 10.0 | 4.5 |

Columns 9–13 of 13 · first column repeated

| | Stock_Available | Payment_Method | Total_Sales_241UDS0103 | Discount_Amount_241UDS0103 | Net_Sales_241UDS0103 |
|---|---|---|---|---|---|
| **Item_ID** | | | | | |
| **E222** | 52 | Cash | 13440 | 672.0 | 12768.0 |
| **E226** | 33 | Cash | 22230 | 1778.4 | 20451.6 |
| **E209** | 29 | UPI | 23520 | 2352.0 | 21168.0 |
| **E214** | 34 | Cash | 25380 | 3299.4 | 22080.6 |
| **E230** | 40 | Card | 37620 | 3762.0 | 33858.0 |

In [ ]:
```
### Step 23 — Sorting Using Multiple Columns
```

In [23]:
```
df_241UDS0103.sort_values(["Category", "Units_Sold"], ascending=[True, False]).head()
```

Out[23]: Columns 1–8 of 13

| Item_ID | Item_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent | Customer_Rating |
|---|---|---|---|---|---|---|---|
| **E226** | Desk Mat | Accessories | Vijayawada | 57 | 390 | 8.0 | 4.1 |
| **E224** | Travel Organizer | Accessories | Delhi | 45 | 690 | 9.0 | 4.5 |
| **E213** | Laptop Stand | Accessories | Chennai | 31 | 1150 | 6.0 | 4.6 |
| **E206** | LED Desk Light | Electronics | Mumbai | 72 | 680 | 12.0 | 4.1 |
| **E230** | USB Hub | Electronics | Bengaluru | 66 | 570 | 10.0 | 4.5 |

Columns 9–13 of 13 · first column repeated

| Item_ID | Stock_Available | Payment_Method | Total_Sales_241UDS0103 | Discount_Amount_241UDS0103 | Net_Sales_241UDS0103 |
|---|---|---|---|---|---|
| **E226** | 33 | Cash | 22230 | 1778.4 | 20451.6 |
| **E224** | 23 | Cash | 31050 | 2794.5 | 28255.5 |
| **E213** | 18 | Card | 35650 | 2139.0 | 33511.0 |
| **E206** | 43 | Cash | 48960 | 5875.2 | 43084.8 |
| **E230** | 40 | Card | 37620 | 3762.0 | 33858.0 |

In [ ]:
```
### Step 24 – Updating Data in the Table
```

In [24]:
```
df_241UDS0103["City"] = df_241UDS0103["City"].replace("Hyderabad", "Hyd")
df_241UDS0103[df_241UDS0103["City"] == "Hyd"].head(2)
```

Out[24]: Columns 1–8 of 13

| Item_ID | Item_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent | Customer_Rating |
|---|---|---|---|---|---|---|---|
| E204 | Portable Charger | Electronics | Hyd | 63 | 1250 | 7.0 | 4.5 |
| E209 | Reusable Bottle | Lifestyle | Hyd | 56 | 420 | 10.0 | 4.8 |

Columns 9–13 of 13 · first column repeated

| Item_ID | Stock_Available | Payment_Method | Total_Sales_241UDS0103 | Discount_Amount_241UDS0103 | Net_Sales_241UDS0103 |
|---|---|---|---|---|---|
| E204 | 37 | UPI | 78750 | 5512.5 | 73237.5 |
| E209 | 29 | UPI | 23520 | 2352.0 | 21168.0 |

In [ ]:
```
### Step 25 – Updating a Specific Product Value.
```

In [25]:
```
df_241UDS0103.loc["E201", "Unit_Price"] = 900
print(df_241UDS0103.loc["E201", "Unit_Price"])
```

```
900
```

In [ ]:
```
### Step 26 – Changing Column Names
```

In [26]:
```
df_241UDS0103 = df_241UDS0103.rename(columns={"Item_Name": "Product",
"City": "Location", "Customer_Rating": "Rating", "Net_Sales_241UDS0103":
"Revenue"})
df_241UDS0103.head(1)
```

Out[26]: Columns 1–8 of 13

| Item_ID | Product | Category | Location | Units_Sold | Unit_Price | Discount_Percent | Rating |
|---|---|---|---|---|---|---|---|
| E201 | Solar Lantern | Renewable Energy | Vijayawada | 38 | 900 | 5.0 | 4.4 |

Columns 9–13 of 13 · first column repeated

| Item_ID | Stock_Available | Payment_Method | Total_Sales_241UDS0103 | Discount_Amount_241UDS0103 | Revenue |
|---|---|---|---|---|---|
| E201 | 24 | UPI | 32300 | 1615.0 | 30685.0 |

In [ ]:
```
### Step 27 — Final DataFrame
```

In [27]:
```
df_241UDS0103.head()
```

Out[27]: Columns 1–8 of 13

| Item_ID | Product | Category | Location | Units_Sold | Unit_Price | Discount_Percent | Rating |
|---|---|---|---|---|---|---|---|
| **E201** | Solar Lantern | Renewable Energy | Vijayawada | 38 | 900 | 5.0 | 4.4 |
| **E202** | Smart Thermostat | Smart Home | Chennai | 27 | 2400 | 8.0 | 4.6 |
| **E203** | Water Purifier | Home Appliance | Visakhapatnam | 44 | 3200 | 10.0 | 4.2 |
| **E204** | Portable Charger | Electronics | Hyd | 63 | 1250 | 7.0 | 4.5 |
| **E205** | Air Quality Sensor | Smart Home | Pune | 21 | 1850 | 6.0 | 4.7 |

Columns 9–13 of 13 · first column repeated

| Item_ID | Stock_Available | Payment_Method | Total_Sales_241UDS0103 | Discount_Amount_241UDS0103 | Revenue |
|---|---|---|---|---|---|
| **E201** | 24 | UPI | 32300 | 1615.0 | 30685.0 |
| **E202** | 15 | Card | 64800 | 5184.0 | 59616.0 |
| **E203** | 19 | Cash | 140800 | 14080.0 | 126720.0 |
| **E204** | 37 | UPI | 78750 | 5512.5 | 73237.5 |
| **E205** | 12 | Card | 38850 | 2331.0 | 36519.0 |

In [ ]:
```
### Step 28 — Final Dataset Information
```

In [28]:
```
df_241UDS0103.info()
```

```
<class 'pandas.DataFrame'>
Index: 30 entries, E201 to E230
Data columns (total 12 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   Product                     30 non-null     str
 1   Category                    30 non-null     str
 2   Location                    30 non-null     str
 3   Units_Sold                  30 non-null     int64
 4   Unit_Price                  30 non-null     int64
 5   Discount_Percent            30 non-null     float64
 6   Rating                      30 non-null     float64
 7   Stock_Available             30 non-null     int64
 8   Payment_Method              30 non-null     str
 9   Total_Sales_241UDS0103      30 non-null     int64
 10  Discount_Amount_241UDS0103  30 non-null     float64
 11  Revenue                     30 non-null     float64
dtypes: float64(4), int64(4), str(4)
memory usage: 4.1+ KB
```

In [ ]:
```
### RESULT
Pandas data manipulation operations were performed on the eco-retail
dataset, including selecting rows and columns, creating calculated fields,
filtering, sorting, updating values, and renaming columns.
```